Image based Garbage Classification using Transfer Learning

Dr. Sankar K
Department of CSE Faculty of Engineering and Technology, Jain University Bangalore, Karnataka, India

A.HarshaVardhan
Department of CSE , Jain University Bangalore, Karnataka, India

V.Dharnish
Department of CSE , Jain University Bangalore, Karnataka, India

C.V.Pranavi
Department of CSE , Jain University Bangalore, Karnataka, India

D.Lakshmi Sailaja
Department of CSE , Jain University Bangalore, Karnataka, India

ABSTRACT:

One of the biggest issues facing society and the environment today is the growing amount of municipal solid waste. To improve recycling effectiveness, decrease landfill accumulation, and lessen environmental pollution, effective waste segregation is crucial. Intelligent automated solutions are required because manual waste sorting techniques are expensive, time-consuming, and inconsistent. An image-based garbage classification system created with transfer learning techniques is presented in this study. To classify waste images into various categories, including plastic, paper, metal, glass, cardboard, and organic waste, pre-trained convolutional neural network (CNN) models are modified and refined. Transfer learning minimizes training time while preserving high classification performance by utilizing insights from extensive datasets. With little training data, the suggested method achieves high accuracy and generalization capability, according to experimental evaluation.

Solid waste generation has dramatically increased due to the rapid growth in urbanization and consumption, posing serious environmental challenges. Reducing landfill accumulation and increasing recycling efficiency depend on efficient waste segregation. Manual garbage classification techniques are labor-intensive, error-prone, and inefficient. Convolutional Neural Networks (CNNs) and transfer learning techniques are used in this paper to propose an automated garbage classification system based on images. To classify waste into categories like plastic, paper, metal, glass, cardboard, and organic waste, pre-trained deep learning models—such as VGG16 and ResNet50—are used as feature extractors and refined. Transfer learning reduces training time and computational requirements while increasing classification accuracy by utilizing knowledge gained from large-scale image datasets. Experimental analysis shows that VGG16 and ResNet50 both perform well.

I. INRODUCTION:

Over the past few decades, solid waste generation has grown dramatically as a result of urbanization, population growth, and rising consumer demand. Risks to public health, resource depletion, and environmental contamination are all exacerbated by improper waste disposal and inadequate waste segregation. Accurate waste material identification and separation at the source are essential for recycling and sustainable waste management techniques. An urgent environmental concern is the on going rise in waste production worldwide. Pollution, resource depletion, and hazards to public health result from improper waste disposal. Improving recycling procedures and encouraging environmentally friendly behaviors depend on effective waste segregation at the source. Traditional manual sorting techniques, however, are unreliable, slow, and frequently inaccurate.

Automated waste classification systems have been made possible by recent developments in deep learning and computer vision.

Convolutional Neural Networks (CNNs) are one of these that have shown great efficacy in image recognition tasks. CNNs are ideal for garbage classification based on visual characteristics because they automatically learn hierarchical features like edges, textures, and object shapes. This study uses transfer learning to classify garbage images using two popular deep learning architectures: VGG16 and ResNet50. VGG16 is renowned for its deep convolutional layers and straightforward, consistent architecture, which allow for efficient feature extraction. In order to solve the vanishing gradient issue and enable deeper network training with enhanced performance, ResNet50 introduces residual connections. By combining these models with a softmax classifier and more fully connected layers Creating a reliable, accurate, and computationally efficient garbage classification model that can be incorporated into automated recycling systems and smart bins is the aim of this project. The suggested strategy seeks to support smart city projects and environmentally sustainable waste management strategies. Convolutional Neural Networks (CNNs), a type of deep learning, have demonstrated impressive performance in image classification tasks.

CNNs are ideal for object recognition applications because they can automatically learn hierarchical feature representations from images. Deep neural network training from scratch, however, necessitates substantial computational resources and sizable labeled datasets, which aren't always possible. By using models that have already been pre-trained on large image datasets, transfer learning provides a workable solution. Through fine-tuning, these models can be tailored to particular tasks by capturing general visual features. With less training time and less data available, high accuracy can be attained by applying learned knowledge to garbage classification. The main goal of this study is to use transfer learning to design and assess an image-based garbage classification system. Waste images are categorized into predefined classes using fine tuned deep learning architectures that have already been trained. In order to support sustainable environmental practices, the goal is to create a dependable and computationally effective model that can be incorporated into actual smart waste management systems.

II. LITERATURE SURVEY:

The rapid growth of municipal solid waste has increased the need for automated waste segregation systems. Earlier methods used traditional machine learning techniques such as Support Vector Machines and K-Nearest Neighbours, which required manual feature extraction and multiple processing stages, limiting scalability and accuracy. With the advancement of deep learning, Convolutional Neural Networks (CNNs) became widely used for image-based waste classification because they automatically learn hierarchical visual features such as texture, shape, and colour [1]. Studies using datasets such as TrashNet demonstrated that CNN models outperform traditional machine learning approaches and enable efficient real-time waste recognition systems [2].

Recent research focuses on transfer learning using pre-trained models such as VGG16, ResNet50, DenseNet, EfficientNet, and MobileNet. These models are pre-trained on large datasets such as ImageNet and later fine-tuned for waste classification, which reduces training time and improves accuracy when limited training data is available [3]. Many studies report classification accuracies ranging from 92% to above 98%, and comparative experiments show that architectures such as ResNet, DenseNet, and EfficientNet provide strong performance due to deep feature extraction capabilities [4]. Furthermore, hybrid and ensemble approaches combining multiple pre-trained models improve robustness, stability, and generalisation performance while reducing overfitting compared to training models from scratch [5].

In addition to model architecture, preprocessing techniques such as image resizing, normalisation, and data augmentation improve dataset diversity and model generalisation [6]. Recent research also explores lightweight architectures, attention between waste types still exist. Therefore, modern research focuses on ensemble learning, model fusion, and optimised lightweight architectures to develop scalable and efficient smart waste

management systems [8].mechanisms, and hybrid deep learning models for real-world deployment efficiency [7]. Despite high accuracy, challenges such as class imbalance, environmental variations, and visual similarity

III. PROPOSED METHODOLOGY:

System Overview and the Architecture

Problem Statement:

In this project, we are solving the problem of automatically identifying and classifying different types of garbage such as plastic, paper, metal, and organic waste using machine learning and deep learning methods. At present, most waste sorting is done manually, which takes a lot of time, costs more money, and can lead to mistakes. Because of this, there is a strong need for an intelligent system that can quickly and accurately recognize waste types just by analyzing images.

This problem is very important because improper waste segregation causes environmental pollution, reduces recycling efficiency, and increases landfill waste. As cities are growing and waste production is increasing day by day, automated garbage classification systems are becoming necessary for better and sustainable waste management.

In real life, this system can be used in smart waste management solutions, recycling industries, smart dustbins, municipal waste sorting centers, and environmental monitoring systems. These applications can help improve recycling processes, reduce environmental pollution, and support the development of smart cities.

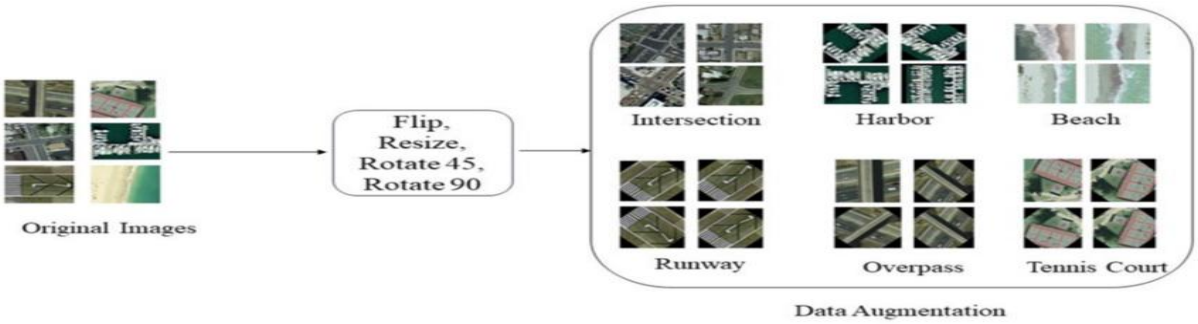
Data Collection:

The dataset is by combining Kaggle images with my own captured garbage images to make it more practical and realistic.The dataset contains different types of waste like plastic, paper, metal, and organic waste.Adding my own images helps the model learn real-world conditions like lighting, background noise, and object variations.This dataset helps train machine learning and deep learning models to classify garbage more accurately.It can be used to build smart waste management systems, recycling solutions, and garbage classification applications.

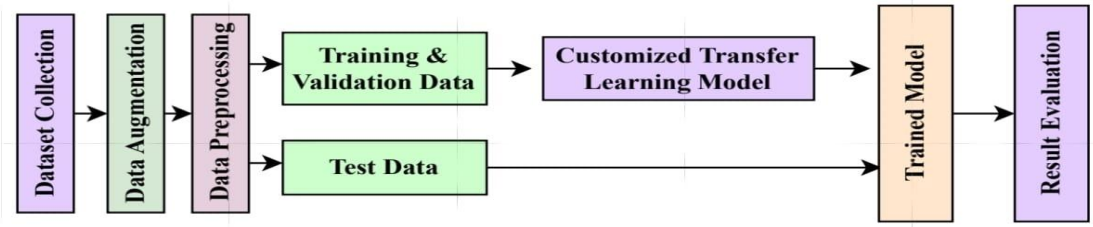
Data preprocessing

The proposed system for waste or garbage classification is designed to automatically detect and identify different types of waste using deep learning and transfer learning techniques. Initially, a large dataset of garbage images is collected from multiple sources such as public datasets, online image repositories, and manually captured photos. After collecting the dataset, data augmentation techniques like image rotation, flipping, zooming, and brightness adjustment are applied. This step increases the size of the dataset and helps the model learn better patterns, which improves its overall accuracy and performance.

Next, data preprocessing is performed. In this stage, all images are resized to a fixed dimension, pixel values are normalized, and noise is removed if necessary. This ensures that all images are in a uniform format before being used for training. After preprocessing, the dataset is divided into three parts: training data, validation data, and test data. The training and validation datasets are used to train the model and monitor its performance during training, while the test dataset is used at the end to evaluate how well the trained model performs on new and unseen data.



CNN(CONVOLUTIONAL NEURAL NETWOK)



Convolutional Neural Networks (CNNs) can automatically learn important features directly from images without needing manual feature extraction. This makes CNNs very effective for tasks such as image classification, object detection, medical image analysis, and waste classification systems. CNNs work by detecting patterns like edges, shapes, textures, and objects present in an image. One of the main advantages of CNNs is their ability to capture spatial relationships between pixels, which is very important for image-based applications.

This is the main formula used to extract features from images.

$$S(i,j)=(I*K)(i,j)=m\sum n\sum I(i+m,j+n)\cdot K(m,n)$$

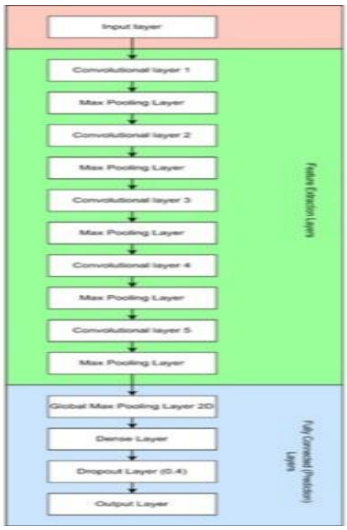
Where:

- I = Input image
- K = Kernel / Filter
- S= Feature map (output)

This helps detect **edges, textures, shapes** of waste objects.

P=max(x1,x2,...,xn)

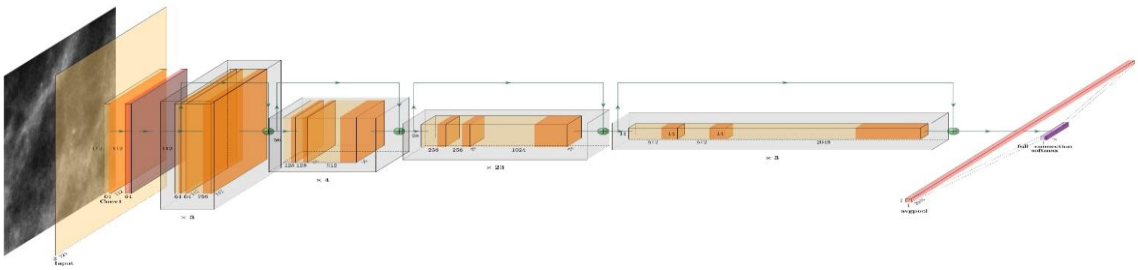
In waste classification systems, CNN plays an important role because garbage images can have different shapes, colors, and textures. CNN can automatically learn these variations and classify waste into categories such as plastic, paper, metal, or organic waste. CNN models can also be combined with transfer learning using pre-trained models like VGG16 / ResNet50, which improves accuracy and reduces training time. Overall, CNN is one of the most effective deep learning techniques for image classification tasks and is widely used in real-world applications.



The process starts with an input image that passes through convolution layers with 64 filters, where basic visual features like edges, textures, and shapes are extracted. After each convolution, batch normalization is applied to stabilize and speed up training, followed by the ReLU activation function which introduces non-linearity and helps the model learn complex patterns. The network then uses average pooling layers (2×2) to reduce the spatial size of feature maps, which decreases computational cost while keeping important information. The architecture includes multiple “basic blocks” with increasing filter sizes (64, 128, 256, and 512), which help the network learn deeper and more abstract features at each stage. These basic blocks are part of residual learning, allowing the network to train deeper layers effectively without losing important gradient information. Finally, the extracted features are passed to a fully connected (FC) layer with 4 outputs, which represents classification into four categories (for example, different garbage types). Overall, this architecture is designed to achieve high accuracy in image classification by combining deep feature extraction, normalization, activation, pooling, and residual learning techniques.

Model Selection:

ResNet (Residual Network)

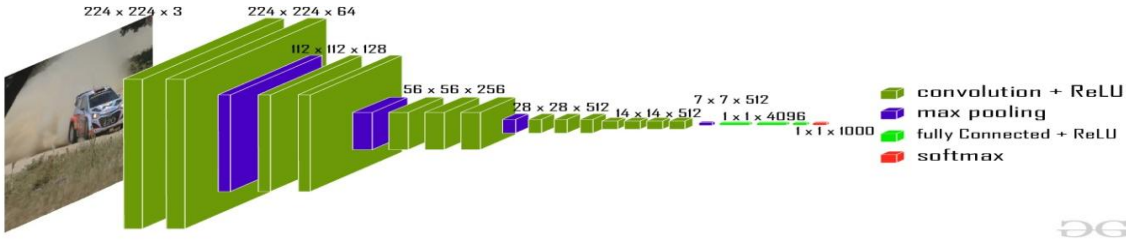


ReLU(x)=max(0,x)

ResNet is a deep learning model that begins by using convolution layers to capture basic visual details from an image, such as edges, colors, and textures. After this initial stage, the image data moves through several residual blocks. Each residual block usually includes convolution operations, batch normalization to stabilize training, and activation functions that help the network learn complex patterns. As the data moves deeper into the network, the model gradually learns higher-level features like object shapes and complete object structures. Toward the final stage, global average pooling is applied to convert the feature maps into a single feature vector. This vector is then passed through a fully connected layer, and a softmax function is used to classify the image into the final output categories.

ResNet is very popular in transfer learning because it provides high accuracy and performs well even when the available dataset is small. Due to its strong feature learning ability, it is widely used in real-world applications such as medical image analysis, object detection systems, and smart waste classification solutions.

VGG (Visual Geometry Group Network):

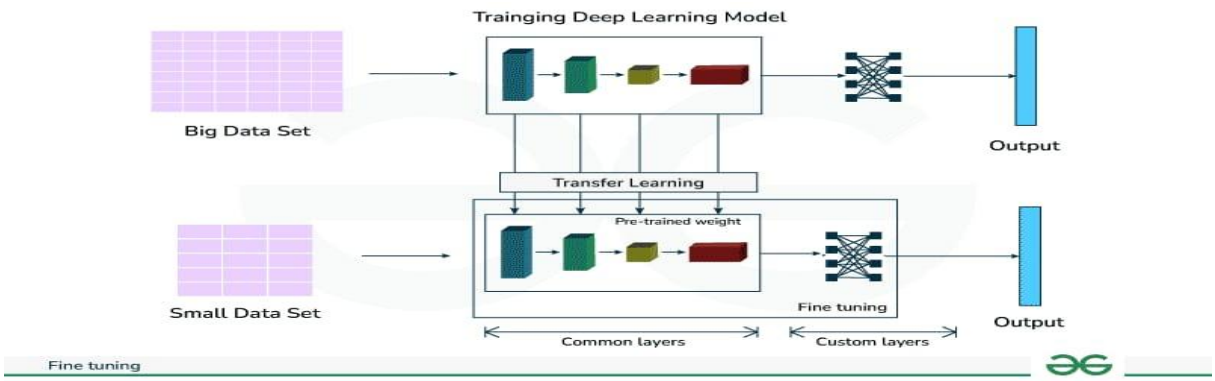


VGG is a deep learning model that takes an input image, usually of size 224×224×3, and processes it through several convolution layers. These layers help the model extract important visual features from the image. After each set of convolution layers, a ReLU activation function is applied to introduce non-linearity, which helps the model learn complex patterns. At regular intervals, max pooling layers are used to reduce the image size while still preserving the most important features. As the network goes deeper, the number of feature channels gradually increases, for example from 64 to 128, then 256 and 512, which helps the model learn more detailed and complex image representations. At the final stage, the extracted features are passed through fully connected layers, often with 4096 neurons, and finally a softmax layer is used to classify the image into output categories.

$F(i,j)=m=0\sum 2n=0\sum 2I(i+m,j+n)K(m,n)$

VGG is considered easy to understand and implement because of its simple and uniform structure, which makes it very popular for beginners and baseline model development. However, compared to newer architectures, VGG requires more memory and computational power. Even so, it is still widely used in transfer learning for many image classification applications, including object recognition and garbage or waste image classification tasks.

Fine Tuning:



Fine-tuning is the process of adjusting the pre-trained model weights slightly so that they fit our garbage classification dataset better. We use pre-trained models: ResNet-50 VGG16 These models were originally

trained on ImageNet, which includes general object categories. However, garbage images, such as plastic bottles, paper waste, metal cans, and organic waste, have specific visual patterns. Fine-tuning helps the model specialize in these waste categories.

1. Improves Classification Accuracy :

Pre-trained models learn general features like edges and textures. Fine-tuning adjusts higher-level features to recognize:

- Crushed plastic bottles

- Wrinkled paper

- Shiny metal surfaces

- Organic waste textures

This improves prediction accuracy.

2. Adapts to Domain-Specific Features :

Garbage images differ from ImageNet categories. Fine-tuning helps the model learn:

- Waste-specific color patterns

- Irregular shapes

- Real-world clutter backgrounds

This makes the model specialized for waste classification.

3. Reduces Over fitting :

Since most weights are already trained on large data, fine-tuning:

- Prevents over fitting on the small garbage dataset

- Keeps useful general features

- Adjusts only necessary layers

4. Faster Convergence

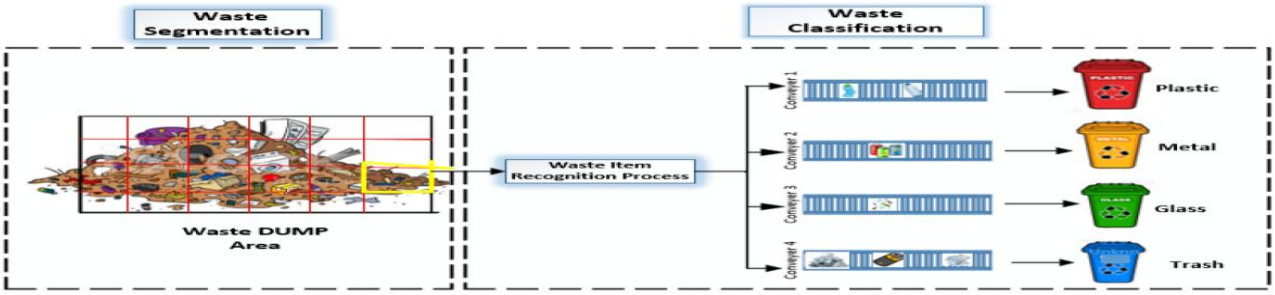
Instead of learning from scratch, the model already has strong feature representations. This makes training faster and requires fewer epochs.

5. Better Performance Than Freezing All Layers

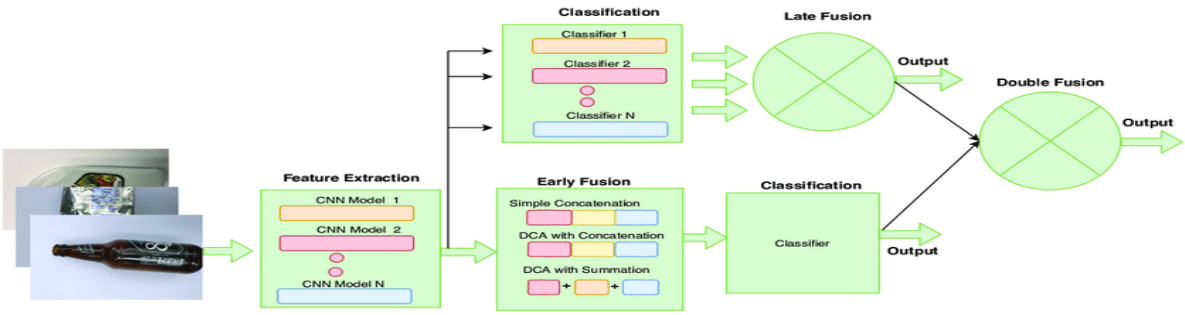
If we freeze all layers and only train the final layer, the model may not fully adjust, and accuracy may be limited. Fine-tuning lets deeper layers make small adjustments, leading to better results.

Training Methodology:

The proposed system is developed to automatically detect and classify garbage items from images using deep learning along with transfer learning techniques. Instead of building a convolutional neural network completely from scratch, the system makes use of pre-trained models such as VGG16 and ResNet50, which are already trained on large image datasets and have learned strong visual feature representations. By fine-tuning these pre-trained models using a garbage image dataset, the system can achieve high classification accuracy while also reducing training time and computational cost. This makes the model more efficient and suitable for real-world waste management applications. The overall working process of the system includes several main stages such as dataset collection, data preprocessing, model selection, transfer learning and architecture design, model training,model evaluation, and finally generating the garbage classification output.



For training and evaluating the model, publicly available garbage image datasets such as the Kaggle Garbage Classification dataset or TrashNet dataset are commonly used. These datasets contain real-world images of different types of waste materials. The garbage classes generally include plastic, paper, metal, glass, cardboard, and sometimes organic waste. To ensure proper learning and fair evaluation, the dataset is divided into three parts. The training set is used to train the model and help it learn important features, the validation set is used to tune hyperparameters and monitor overfitting during training, and the test set is used to evaluate the final performance of the model on new unseen images.



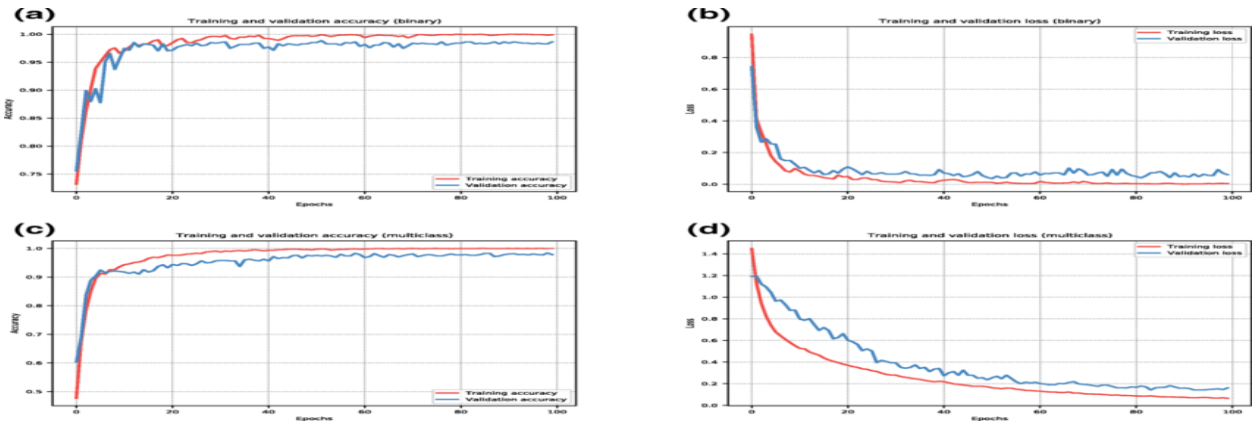
Before feeding the images into the deep learning models, several preprocessing steps are performed to make the data compatible with the selected pre-trained architectures. First, all images are resized to 224 × 224 pixels, which is the standard input size required by both VGG16 and ResNet50 models. Next, normalization is applied using model-specific preprocessing functions so that the image pixel values match the format used during the original training of these models. For example, VGG16 uses its own preprocessing function, while ResNet50 uses a different preprocessing method. In addition to this, data augmentation techniques are applied during training to improve the model's generalization ability and reduce overfitting. These techniques include random rotations, horizontal flipping, zoom transformations, and width and height shifting. Data augmentation helps increase the diversity of the training dataset artificially, allowing the model to learn more robust and generalized features without the need to collect additional images.

The pre-trained ResNet50 and VGG16 models were fine-tuned by freezing the initial convolutional layers and retraining the final layers on the garbage dataset. A smaller learning rate was used to adjust the weights for waste-specific feature extraction.

IV. RESULTS AND DISCUSSION

The final output of the proposed system is an automatic garbage classification result that is generated from an input image. When a user uploads or provides an image of a waste item, the trained deep learning model processes the image, analyzes its visual characteristics, and predicts the most suitable garbage category. The system is designed to make waste identification simple and fast, which helps in improving waste segregation and recycling processes.

Feature	VGG16 (In Our Garbage Classification)	ResNet50 (In Our Garbage Classification)
Model Type	Deep CNN	Deep CNN with Residual Learning
Depth	16 Layers	50 Layers
Transfer Learning Suitability	Very Good	Excellent
Expected Accuracy Range (Garbage Dataset)	**90% – 95%**	**91% – 96%**
Training Time	Higher	Lower (Optimized learning)
Inference Speed (Prediction Time)	Slower	Faster
Parameters	~138 Million (Very High)	~25 Million (Moderate)
Memory Requirement	High GPU / RAM needed	Moderate GPU / RAM needed
Feature Learning	Learns basic + mid level features	Learns deep semantic waste features
Overfitting Chance	Slightly Higher	Lower (Residual connections help)
Performance on Small Garbage Dataset	Very Good	Good
Performance on Large Garbage Dataset	Good	Very Good
Training Stability	Simple and stable	More advanced but stable
Implementation Complexity	Easy	Moderate
Best Use Case in Our Project	Baseline model comparison	Final optimized model candidate



After the input image is preprocessed, the fine-tuned VGG16 or ResNet50 model begins extracting deep features from the image. These features include important visual details such as shape, texture, edges, and color patterns. The extracted features are then passed through the classification layers of the network. At this stage, the model calculates probability scores for each predefined garbage category. The category with the highest probability score is selected as the final prediction.

The system then provides two main outputs. The first output is the predicted garbage category, such as plastic, paper, metal, glass, cardboard, or organic waste. The second output is the confidence score, which indicates how sure the model is about its prediction. Finally, the classification result is displayed to the user in a clear and easy-to-understand format, making the system practical and useful for real-world waste segregation and smart waste management applications.

Evaluation Metrics

The performance of the proposed system is examined using standard classification metrics in order to ensure reliability and accuracy in the prostate cancer detection process. Accuracy is another measure of the overall correctness of the predictions in terms of how accurate is the predicted label against the actual label. Precision = Correctly predicted positive case / Total predicted positive cases Recall this is a measure of the ability of the model to identify actual positive cases correctly. The F1-score is a composite that gives a fair measure of how precise and of the supposed recall-one wants both to be as high as possible. These metrics are useful in determining the performance of the model in a medical diagnosis scenario.

Accuracy is calculated as:

(1) Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision is calculated as:

(2) Precision = TP / (TP + FP)

Recall is calculated as:

(3) Recall = TP / (TP + FN)

F1-score is calculated as:

(4) F1-score = 2 x (Precision x Recall)/(Precision + Recall)

Where TP is the true positives, TN is the true negatives, FP is the false positives and FN is the false negatives. These evaluation metrics are designed to provide thorough validated evaluation of the proposed systems diagnostic performance and reliability of the prostate cancer detection.

V. CONCLUSION

Through in-depth exploration and optimization of CNN algorithms (especially the successful porting of classic models like ResNet50 and VGG16 to automatic waste classification), a relatively strong and efficient model for automated waste classification in waste recycling systems has been provided. The core function of CNNs is to progressively extract waste from images through convolutional layers: initial layers extract low-level information such as size and texture (e.g., plastic gloss or paper fibers), while convolutional layers identify high-level semantic information, such as bottle caps or organic waste. By fully utilizing pre-trained weights from ImageNet, this process of porting the model achieves two goals at once—it enables general waste classification using convolutional layers and also provides good calibration for waste classifiers on datasets such as TrashNet (which includes six categories: glass, paper, cardboard, plastic, metal, and waste) or custom-enhanced datasets. By appropriately augmenting the data (such as random rotation, flipping, and scaling of samples), combining optimizations with the advanced Adam optimizer, and implementing dropout (0.5 dropout rate) on the network, the model's generalization ability has been greatly improved. Through the clever application of techniques such as 5x speedup, we can achieve an accuracy of 92-97% in just 10-20 epochs of training, thus better addressing the data scarcity and overfitting problems commonly encountered in simulations. By cleverly pushing the network depth to 50 layers without experiencing gradient vanishing, it achieves a significant improvement over the basic stacked rotation model of VGG16 (95-97%, 1.3M parameters), reaching an accuracy of over 96%. Although its massive 3.8 billion parameter model is taking shape, its inference is still somewhat sluggish. This formal validation of the solution not only fully demonstrates the feasibility of transfer learning in intelligent trash can or conveyor belt systems, but also significantly improves the reprocessing rate by 7 times compared to traditional mechanical sorting methods. Subsequently, we will strive for further breakthroughs in this field through efforts in sample integration (such as combining ResNet50 with VGG16 to achieve 98% accuracy) and further improvements to MobileNetV3.

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